

State Huygens principle. Using it draw a diagram showing the details of passage of a plane wave from a denser into a rarer medium.

Draw a ray diagram to show the formation of an image at the least distance of distinct vision, by a compound microscope. Hence, obtain an expression for its angular magnification.

With the help of a circuit diagram, describe the method to obtain p-n junction diode characteristic when it is forward biased. Draw the shape of the characteristic.

When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady? Why?

Electrostatics

1. State and explain Gauss's Law.
2. Electric field due to an electric dipole (axial position).
3. Electric field due to an electric dipole (equatorial position).
4. Potential due to an electric dipole.
5. Capacitors in series and parallel.
6. Effect of dielectric on capacitance.

Current Electricity

7. Drift velocity and its relation with current.
8. Kirchhoff's Laws and their applications.
9. Meter Bridge – principle and numerical.
10. Potentiometer – comparison of EMFs.
11. Determination of internal resistance using potentiometer.

Moving Charges & Magnetism

12. Biot-Savart Law.
13. Ampere's Circuital Law.
14. Magnetic field due to a long solenoid.
15. Force between two parallel current-carrying conductors.
16. Cyclotron – construction and working.

Using Gauss's theorem, obtain an expression for the electric field intensity due to an infinitely long straight uniformly charged wire.

Draw a ray diagram of compound microscope.

Derive expression for magnifying power when final image is formed at least distance of distinct vision.

- (i) How will the interference pattern in Young's double slit experiment be affected if the monochromatic source is replaced by a source of white light ?
- (ii) In Young's double slit experiment, the two slits are 0.18 mm apart and the screen is placed 1.4 m away from the plane of slits. If the distance between the central bright fringe and the third bright fringe is 1.4 cm , then calculate the wavelength of light used.

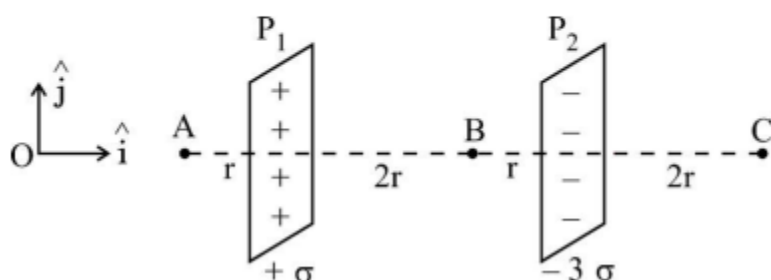
Write two differences between matter waves and electromagnetic waves.

Find the ratio of de Broglie wavelengths associated with deuterium and tritium if both have the same kinetic energies.

In photoelectric effect experiment, show the variation of

- (a) photocurrent with collector plate potential for a given surface for different intensities of incident radiation. Do the curves meet at any point ? If so, why ?
- (b) photocurrent with intensity of radiation incident on a surface keeping the frequency and plate potential fixed.

Two large plane sheets P_1 and P_2 having charge densities $+\sigma$ and -3σ respectively are arranged parallel to each other as shown in the figure. Find the net electric field ($\vec{E}$) at points A, B and C.



Optical fibres are very thin fibres made of glass or plastic and are used in the field of communication. They are flexible, light in weight and can transmit signals over long distances with very small loss of energy. An optical fibre consists of an inner core of refractive index n_1 surrounded by a cladding of refractive index n_2 such that $n_1 > n_2$. When a light ray enters the core at one end, it suffers repeated total internal reflections at the core-cladding interface and finally comes out from the other end. This phenomenon is called total internal reflection. Optical fibres are used in telecommunication, endoscopy and in the field of medicine.

- (i) The phenomenon responsible for the working of optical fibre is

(A) Refraction	(B) Dispersion
(C) Total internal reflection	(D) Interference
- (ii) For total internal reflection to occur, the light must travel from

(A) Rarer to denser medium	(B) Denser to rarer medium
(C) Denser to denser medium	(D) Any medium to any medium
- (iii) If the critical angle for the core-cladding interface is 60° , then total internal reflection will occur when the angle of incidence in the core is

(A) 30°	(B) 45°	(C) 60°	(D) 75°
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- (iv) Which of the following is a correct reason for the use of optical fibres in communication?

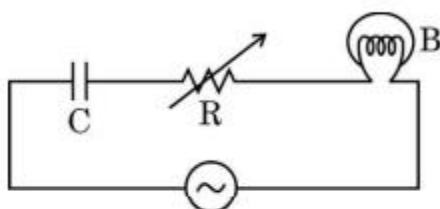
(A) They are heavy and cheap.	(B) They have very high attenuation.
(C) They can transmit light with very small loss of energy.	(D) They require high power for operation.
- (v) The numerical aperture (NA) of an optical fibre is related to the refractive indices n_1 and n_2 as

(A) $NA = \sqrt{n_1^2 - n_2^2}$	(B) $NA = \frac{n_2}{n_1}$
(C) $NA = n_1 + n_2$	(D) $NA = \frac{n_1 - n_2}{n_1 + n_2}$

When light of suitable frequency is incident on a metal surface, electrons are emitted from the surface. This phenomenon is called photoelectric effect. Einstein explained this phenomenon on the basis of quantum theory of light. According to Einstein, light consists of discrete packets of energy called photons. The energy of a photon is given by $E = h\nu$, where h is Planck's constant and ν is the frequency of light. For emission of electrons, the energy of the photon must be at least equal to the work function (ϕ) of the metal. The maximum kinetic energy of emitted electrons is given by $K_{max} = h\nu - \phi$.

- (i) The minimum frequency of incident light required to just emit electrons from a metal surface is called
 - (A) Critical frequency
 - (B) Threshold frequency
 - (C) Cut-off frequency
 - (D) Resonant frequency
- (ii) If the intensity of incident light is increased (keeping frequency constant and above threshold), then
 - (A) Stopping potential increases
 - (B) Number of photoelectrons increases
 - (C) Kinetic energy of photoelectrons decreases
 - (D) Threshold frequency increases
- (iii) The stopping potential for a photosensitive metal depends on
 - (A) Intensity of incident light
 - (B) Frequency of incident light
 - (C) Area of the metal surface
 - (D) Work function of the metal
- (iv) Einstein's photoelectric equation is
 - (A) $E = mc^2$
 - (B) $h\nu = \phi + K_{max}$
 - (C) $V = IR$
 - (D) $F = q(E + \nu \times B)$
- (v) When light of frequency less than the threshold frequency is incident on a metal surface, then
 - (A) Electrons are emitted with less kinetic energy
 - (B) No electrons are emitted
 - (C) Electrons are emitted with more kinetic energy
 - (D) Emission takes place after some time

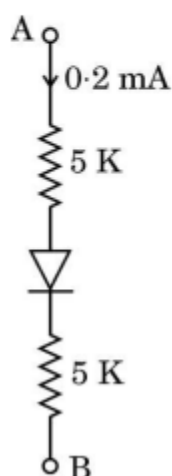
The figure shows a capacitor C , a variable resistor R and a bulb connected in series to the ac source of frequency (ν). The bulb glows with some brightness.



How will the glow of the bulb be affected, if the

- separation between the plates of the capacitor is doubled, keeping resistance R and frequency (ν) the same ?
 - resistance R is decreased keeping the value of capacitance C and frequency (ν) same ?
 - frequency of ac source is decreased for the same value of C and R ?
- Justify your answer in each case.

- Draw the circuit diagram used to study $I - V$ characteristics of a p-n junction diode in conducting mode. Mark on the graph the threshold voltage of the diode. Explain the significance of this voltage.
- In the circuit shown in the figure, the forward voltage drop across the diode is 0.3 V . Find the voltage difference between A and B.



Briefly describe the classification of solids into metals, insulators and semi-conductors on the basis of energy level diagrams.

In a silicon diode, the current increases from 10 mA to 20 mA when the voltage changes from 0.6 V to 0.7 V .

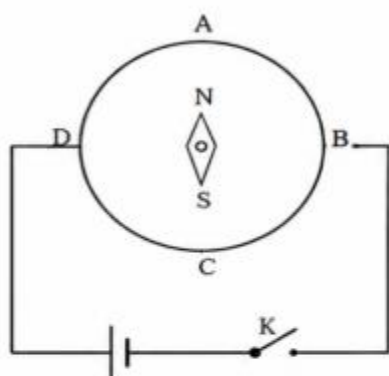
Calculate the dynamic resistance of the diode.

Why does intensity not affect stopping potential?

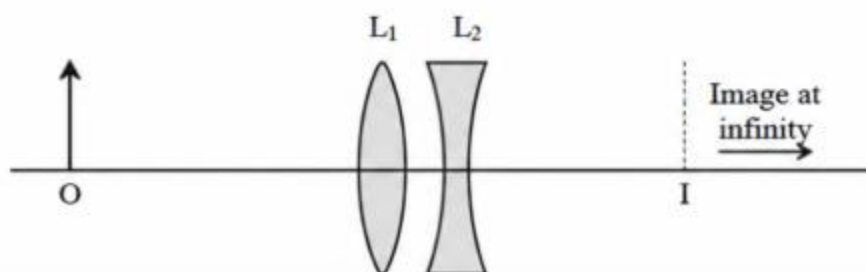
Name the phenomenon responsible for transmission.

Define critical angle.

- (i) A small magnetic needle placed at the centre of a circular current carrying loop ABCD remains in the plane of the loop. Show that when the key K is closed, the needle deflects and at right angles to the plane of the loop. State the factors on which the angle of deflection depends.



- (ii) A convex lens L_1 of focal length 20.0 cm is placed coaxially in contact with a concave lens L_2 of focal length 30.0 cm. Where should an object be placed on the principal axis so that the final image is formed at infinity? Draw a ray diagram to justify your answer.

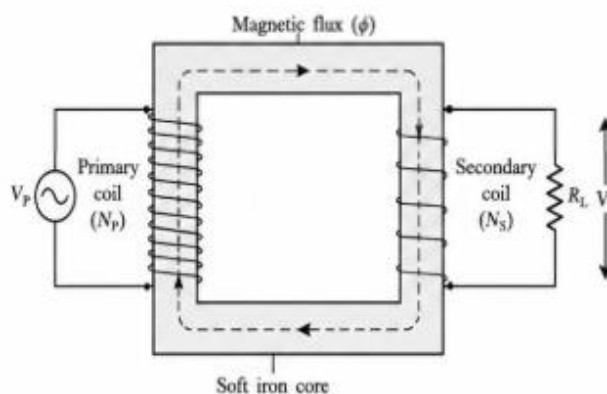


- (i) With the help of a labelled diagram, explain the construction and working of a step-down transformer. State its principle.

- (ii) A transformer has 2000 turns in the primary coil and 200 turns in the secondary coil. If the input AC voltage is 220 V and the current in the primary coil is 0.5 A, calculate:

- (a) the output voltage
(b) the current in the secondary coil
(c) the power output

Assuming the transformer to be ideal.

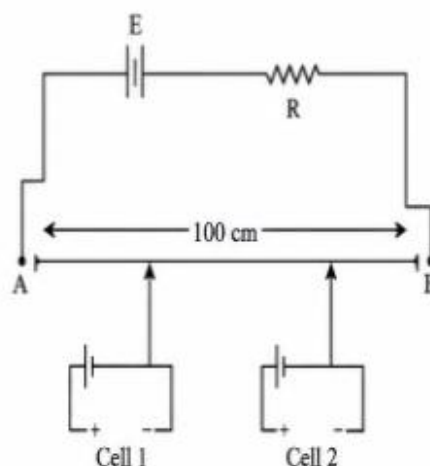


- (i) Explain the working of a potentiometer to compare the emf of two cells.

Draw a labelled circuit diagram for the same.

- (ii) In the circuit shown in the figure, the balancing length obtained in the potentiometer experiment for cell 1 is 63.0 cm and for cell 2 is 81.0 cm. If the emf of cell 1 is 1.26 V, calculate the emf of cell 2.

Why is the potentiometer preferred over a voltmeter for measuring emf of a cell?



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